

Suvarnsing Bhable

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- OBJECTIVE:** *Looking for Results-oriented and challenging career assignments in Automatic Speech Recognition, Natural Language Processing, Remote Sensing and GIS, Machine Learning, and Deep Learning. As a machine learning researcher, I have 6 years and 1 months of experience in **Research and Development** with Python. I know Data Science and cloud platforms to translate my passion experience into effective and outstanding solutions, that could help organizations make informed decisions and gain an edge over their competitors.*
- PROFILE SUMMARY:** Ph.D. Research Scholar (**Mahatma Jyotiba Phule Research Fellowship (MJPRF)**) skilled in Automatic Speech Recognition, Natural Language Processing, and Image Processing with 5 years of relevant experience creating machine learning models and retraining systems and transforming data science prototypes to production-grade solutions. Consistently optimizes and improves real-time recommendation systems by evaluating strategies and testing changes in Python. Consistently employs statistical methods and designs to yield real gains from model changes.
- EXPERIENCE:** **Ph.D in Computer Science** **2018 to 2024.**
Multilingual Acoustic Modeling of Automatic Speech Recognition (ASR) for Indian Languages., I am pursuing a Ph.D. in Computer Science at the Department of Computer Science and Information Technology, Dr. Babasaheb Ambedkar Marathwada University, Aurangabad.
- M.Phil** **2014 to 2018**
We have designed a **Multimodal Biometric System for Face and Ear using Convolutional Neural Network** Convolutional Neural Network (CNN)-based framework Image recognition system. I worked as a Machine Learning Research Student at the Department of Computer Science and Information Technology, Aurangabad, Maharashtra, India.
- EDUCATION:** **Ph.D in Computer Science(Machine Learning)**
Department of Computer Science and Information Technology
Dr.Babasaheb Ambedkar Marathwada University, Aurangabad, Maharashtra, India.
Pursuing.
- Master of Philosophy in Computer Science (M.Phil)**
Department of Computer Science and Information Technology
Dr.Babasaheb Ambedkar Marathwada University, Aurangabad, Maharashtra, India.
2018.
- PG Diploma in Digital and Cyber Forensic and Related Law**
GOVERNMENT INSTITUTE OF FORENSIC SCIENCE Nipatniranjan Nagar, Caves Road, Aurangabad-431004 Maharashtra, India.
2015.
- Master of Science in Computer Science (M.Sc)**
Department of Computer Science Information Technology
Dr.Babasaheb Ambedkar Marathwada University, Aurangabad, Maharashtra, India.
2011-2013.
- Bachelor of Science (B.Sc)**
Indraraj College of Science Sillod, Aurangabad, Maharashtra, India.

2008-2011.

Higher Secondary Education Board (H.S.C 12th)
Shri Shivaji College of Science Buldhana, Maharashtra
2008.

Secondary Education Board (S.S.C 10th)
Shri Chakradhar Swami Vidhyalay Madh, Buldhana, Maharashtra
2006.

PROJECTS:
Project #1

“Multilingual Acoustic Modeling of Automatic Speech Recognition (ASR) for Indian Languages.”
June 2018 to on going

Description :

Speech technology is based on machine learning, deep learning, and statistical models. To develop a system for a new language we need some considerable resources. Multi-lingual acoustic modeling using the Subspace Gaussian Mixture Model for low-resource languages of Indian origin is investigated. We will build an acoustic model(Using Kaldi toolkit) for a low-resource language with limited vocabulary by leveraging resources from another language with comparatively larger resources was focused upon. Experiments will be doing on Hindi and Marathi corpus from the MANDI IIIT Hyderabad database, with Hindi having greater resources than Marathi. The development of models which can factor language-dependent and language-independent aspects of the speech signal, perhaps exploiting in-variance derived from speech production.

- **Technology/Tools:** Python.

Role.

Responsibilities:

- Involved in requirements gathering and end to end system life cycle
- Collected the Database of Speech
- Composed production-grade code to convert machine learning models into pipelines
- Transformed raw data to conform to assumptions of machine learning algorithm. • Conceptualize and design abstract ideas, built and plans
- Implementing and evaluating artificial intelligence and machine learning algorithms and neural networks

Project #2

Description :

“A Multimodal Biometrics System for Face and Ear using Convolutional Neural Networks(CNNs)”
2014 - 2018.

Deep learning is using for a better feature when constructing a machine learning model; it has contained many hidden layers. Biometric individual character verification is increasing additional consideration. Multimodal biometric system is imperative to relate to extraordinary applications in recognizable proof Systems. Execution proportion of the unimodal biometric System is regularly moving back because of physiological deformities, client mode, and the earth. We develop Face and Ear based Multimodal Biometrics for Human recognition in light of part Faster R-CNN is used for detecting faces and ears after that we used CNNs used to train our own nets and identify faces and ears in our database, For fusion Bayesian decision fusion method to fuse recognition result for improving recognizing rate.

- **Technology/Tools:** Python

Role

Responsibilities:

- Involved in requirements gathering and end to end system life cycle
- Collected the standard Database for Face and Ear
- For Face Ear database we take ten different angle
- This project helped me to gain great experience to work on different modules

Project #3 **“SentiView: A Lexicon Based Approach for Twitter Sentiment Analysis” 2013**

Description : The SentiView system uses a lexicon-based or unigram model for classification. The system is developed using the R platform. The data used in this system is crawled by an R twitter API under the package tweetR. The dataset contains approximately 5000 tweets downloaded using the tweeter API. We have downloaded the tweet for the iPhone tag means this will return the tweet that has text in tweet iPhone from recent to old fashion automatically. For our task, we have downloaded 10000 tweets. We got using SentiView we can say that among the all other algorithm Support Vector The machine gives(SVM) the better result. The main reason for high accuracy compared to previous is due to pre-processing and removal of non-opinion tweets from data, this reduces the search space. Due to lexicon-based, the heart of the system is a word dictionary and it is limited. It considers only the word that has in the dictionary .

• **Technology/Tools:** R

Role • Involved in requirements gathering and end to end system life cycle
Responsibilities: • Download the Database for tweets
 • This project helped me to gain great experience to work on large dictionary

SKILLS **Languages :** Python,R
Database : MySQL,MongoDB
Tools/Framework/Library : RSGIS, QGIS, ArcMap, ArcGIS Pro, Whisper openai, Kaldi toolkit, TensorFlow, Keras, NLTK, Spacy, gensim, SpeechRecognition, Scikit-learn, Numpy, Pandas, Scipy, Matplotlib, Seaborn, AWS.
Machine Learning Algorithms : SVM, Logistic Regression, Linear Regression, kNN, Random Forest, Naive Bayes.
Deep Learning Algorithms : ANN, CNN, RNN, LSTM, Backpropagation, Feed-forward Neural Network.
Platforms and Misc.: Anaconda, Jupyter Notebook, Spyder IDE, Windows XP/W7/W8/W10, Linux
General : Data labeling, Data Modeling Evaluation, Probability and Statistics, Data Structures and Algorithm

CERTIFICATION:• *GitHub for Data Scientists* at linkedin learning.
• *Machine Learning with python: Foundations* at linkedin learning.
• *TensorFlow: Working with NLP* at linkedin learning.
• *Carrer Edge* at Tata Consultancy Services.
• *Deep Learning* at MathWorks.
• *Machine Learning* at MathWorks.
• *Matlab* at Matlab Academy, MathWorks.
• *Python* at Naresh i Technologies, Hyderabad, India.
• *Diploma in Machine learning and Data Science* at Naresh i Technologies, Hyderabad, India.

ADDITIONAL ACTIVITIES: • Attended Workshop on IP Awareness/Training program under National Intellectual Property Awareness Mission, Organized by Intellectual Property Office, India. 23 June 2022.
• Attended Workshop on Intellectual Property Right, Incubation System and Government Support for Aspring Startups/ Entrepreneurs, Organized by Atal Incubation Centre, Dr. Babasahrb Ambedkar Marathwada University, Aurangabad, India. 26 April 2022.
• Attended Deep Dive into Patenting Process, Organized by Dr. G. Y. Pathrikar College of Computer Science and Information Technology, MGM University, Aurangabad Marathwada Accelerator for Growth Incubation Council. 21 October 2021, Au-

rangabad, India.

- Attended Three Days National Level Online Faculty Development Program-II Deogiri College, Aurangabad-2021, India.
- Attended Webinar on Microsoft Cloud Technologies Deogiri College, Aurangabad 2021, India.
- Attended International Conference on **INTERSPEECH**-2018 in Hyderabad, India.
- Participated of Two Day UGC-SAP Nation Seminar on Materials Science, X-ray and Gamma Ray Spectroscopies at organized by Department of Physics, Dr.B.A. Marathwada University, Aurangabad March 29-30-2017.
- One Day Workshop on "LaTeX" organized by Department of CS and IT, Dr.Babasaheb A. Marathwada University, Aurangabad, during 25th June 2016.
- Local Organizer volunteer, University Level Avishkar organized by Dr.B.A. Marathwada University, Aurangabad, during Dec 2014, Dec 2015, Dec 2016.
- Attended in 100 The Indian Science Congress Association at Calcutta University, Koltakata from January 3 to 7,2013.
- participated in One Day State Level Seminar on "Cyber Law and Cyber Security" at Dr. G.Y. Pathrikar College of Computer Science Information Technology, Aurangabad.
- Participated in National Workshop on e-Governance organized by Dr.B.A. Marathwada University, Aurangabad (MS)-India and Directorate of Information Technnology of Marathwada on 30 july 2013.
- Participated in International Conference on Knowledge Engineering organized by Department of computer science and IT, Dr.Babasaheb Ambedkar Marathwada University, Aurangabad, during 29-31 Jan 2011.

PUBLICATIONS:

1. Nilesh Misal, **Suvarnsing Bhable**, Sandip Dilip Gaikwad, Ashish Lahase "Advanced Techniques for Identifying and Classifying Slum Area Using Remote Sensing and Geographic Information Systems (GIS)", HTL JOURNAL Volume 30 (Issue 6): 53 - 57 DOI.org/10.37896/HTL30.6/10906
2. **Suvarnsing Bhable**, Charansing Kayte "Multilingual Acoustic Modeling of Automatic Speech Recognition (ASR) for Indian Languages", Strad Research Volume 11(Issue 2): 254 - 260 <https://stradresearch.org/index.php/volume-11-issue-2-2024/>
3. **Suvarnsing Bhable**, Charansing Kayte "Speech Database Acquisition for Multilingual Acoustic Modeling of Automatic Speech Recognition (ASR) for Indian Languages".High Technology Letters Volume 30(Issue 1) : 348-353, 2024 DOI.org/10.37896/HTL30.1/9976
4. Lahase, A. R., **Bhable, S.**, Maher, S., Nimbhore, S. State of the Art on Speaker Diarization for low resource Languages.High Technology Letters volume 29(issue9):350-356 DOI: 10.37896/HTL29.09/9335
5. Lahase, A., **Bhable, S.**, Deshmukh, R., Nimbhore, S. (2023). Comparative Analysis of Automatic Speaker Diarization Techniques. In: Senjyu, T., So-In, C., Joshi, A. (eds) Smart Trends in Computing and Communications. SmartCom 2023. Lecture Notes in Networks and Systems, vol 650. Springer, Singapore. <https://doi.org/10.1007/978-981-99-0838-761>
6. **Suvarnsing Bhable**, Ratnadeep R. Deshmukh and Charansing Kayte, " Comparative Analysis of Automatic Speech Recognition Techniques. In International Conference on Applications of Machine Intelligence and Data Analytics, ICAMIDA 2022.
7. Maher, S. K., **Bhable, S. G.**, Lahase, A. R., and Nimbhore, S. S. (2022, May). AI and Deep Learning-driven Chatbots: A Comprehensive Analysis and Appli-

- cation Trends. In 2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS) (pp. 994-998). IEEE.
8. Kiran Suryavanshi, **Suvarnsing Bhable** and Charansing Kayte , " A Critical Insight into Automatic Visual Speech Recognition System. In International Conference on Artificial Intelligence and Speech Technology (pp. 3-11). Springer, Cham, November 2021.
 9. **Suvarnsing Bhable**, Ashish Lahase, Santosh Maher. " Automatic Speech Recognition (ASR) of Isolated Words in Hindi low resource Language", International Journal for Research in Applied Science Engineering Technology (IJRASET) ISSN: 2321-9653; IC Value: 45.98; Volume 9 Issue II Feb 2021
 10. Ashish Lahase, Santosh Maher and **Suvarnsing Bhable**, "Twitter Data analysis (Classification) for text using Multinomial Naive Bayes", International Journal of All Research Education and Scientific Methods (IJARESM), Volume 9, Issue 3, March -2021.
 11. **S. G. Bhable** and C. N. Kayte, "Review: Multilingual Acoustic modeling of Automatic Speech Recognition(ASR) for low resource languages," 2020 IEEE International Conference on Advent Trends in Multidisciplinary Research and Innovation (ICATMRI), 2020, pp. 1-4, doi: 10.1109/ICATMRI51801.2020.9398431.
 12. Nagsen Bansod, Siddharth B. Dabhade, Jitendra N. Dongre, K. V. Shende, **Suvarnsing G. Bhable**, S. Maher, S. Thorat, K. Ankushe, S. Tharewal, V. S. Jadhav and K. V. Kale Human Electroencephalographic Biometric Person Recognition System Proceedings of 2nd International Conference on Communication, Computing and Networking, Lecture Notes in Networks and Systems 46, <https://doi.org/10.1007/978-981-13-1217-59>.
 13. Santosh K. Maher, Sumegh Tharewal, Abdul Hannan, **Suvarnsing G. Bhable**, K. V. Kale Review on HRV based Prediction and Detection of Heart Disease, International Journal of Computer Applications(IJCA), ISSN No. (0975 8887, Volume 179-Issue 46 ,June 2018, PP.7-12.
 14. **Suvarnsing G. Bhable**, Sandip D. Mali, Jaypalsing N. Kayte, Sangramsing Kayte Oceanography And Climate Change Using Remote Sensing And Gis International Journal of Remote Sensing and Geoscience (IJRSG), ISSN No: 2319-3484, Volume 5, Issue 1, Jan. 2016,PP.24-29.
 15. **Suvarnsing G. Bhable**, Sangramsing Kayte, Sandip Mali, Jaypalsing N. kayte, Raju Maher A Review Paper on Coastal Hazard Journal of Engineering Research and Applications ISSN: 2248-9622, Vol. 5, Issue 12, (Part - 2) December 2015,PP.83-93.
 16. **Suvarnsing G. Bhable**, Sandip Mali, Ashish Balerao, Sangramsing Kayte Sentiment of Sentence in Tweets: A Review IOSR Journal of Computer Engineering (IOSR-JCE) e-ISSN:2278-0661, p-ISSN:2278-8727, Volume 17, Issue 6, Ver. I (Nov Dec. 2015), PP 157-162, <http://www.iosrjournals.org/iosr-jce/pages/17>.
 17. Sangramsing Kayte, Raju Maher, **Suvarnsing G. Bhable**, Jaypalsing Kayte Automated Detection of Microa- neurysm, Hard Exudates, and Cotton Wool Spots in Retinal fundus Images IOSR Journal of Computer Engineering (IOSR-JCE) e-ISSN: 2278-0661,p-ISSN: 2278-8727, Volume 17, Issue 6, Ver. I (Nov Dec.2015), PP 152-156 <http://www.iosrjournals.org/iosr-jce/pages/17>.
 18. Raju Maher , Sangramsing Kayte, **Suvarnsing G. Bhable** "Automatic Diagnosis of Diabetic Retinopathy Micro aneurysm from Low Contrast Retinal Images using Mathematical Morphology Methods International Journal of Computer Applications (0975 8887) Volume 130 No.6,November2015.

19. **Suvarnsing G. Bhable**, Sangramsing Kayte ,Raju Maher, Jaypalsing Kayte, Dr. Charansing Kayte DNA Bio- metric IOSR Journal of VLSI and Signal Processing (IOSR-JVSP) Volume 5, Issue 5, Ver. I (Sep Oct. 2015), PP 2319.
20. Suvarnsing G. Bhable, **Sangramsing Kayte**, Jaypalsing N. Kayte, Dr. Charansing Kayte "Robust Multimodal Biometrics Recognition: A Review International Journal of Advanced Research in Computer Science and Software Engineering -Volume 5, Issue 10, October-2015(Impact Factor:2.5).
21. **Suvarnsing G. Bhable**, Sangramsing N. Kayte , Jaypalsing N. Kayte , Raju Maher A Review Paper on Multimodal Biometrics System using Fingerprint and Signature International Journal of Computer Applications (0975 8887) Volume 128 No.15, October 2015 (Impact Factor:3.12).
22. Sangramsing Kayte, Raju Maher, **Suvarnsing G. Bhable**, Jaypalsing Kayte Automatic Detection of Non- Proliferative Diabetic Retinopathy Using Fundus Images IOSR Journal of Computer Engineering (IOSR-JCE) e- ISSN:2278-0661, p-ISSN:2278-8727, Volume17, Issue6, Ver.I (NovDec.2015), PP125-130.
23. **Suvarnsing G. Bhable**, Sangramsing Kayte, Jaypalsing Kayte Raju Maher, Dr.R.R.Deshmukh Modification and Climate Change Analysis of surrounding Environment using Remote Sensing and Geographical Information System IOSR Journal of Computer Engineering (IOSR-JCE) e-ISSN:2278-0661, p-ISSN:2278-8727,Volume17, Issue 6, Ver. I (Nov Dec. 2015), PP 138-146.

**PERSONAL
INFORMATION:**

Permanent Address:

At Post Madh Tq.+ Dist. Buldhana,
Maharashtra, India-443002

Gender: Male

Marital Status: Single

Passport Number: M1979936

PAN Number: BRIPB5431J

Nationality: Indian

GitHub: <https://github.com/suvarnsingbhable>

Researchgate: <https://www.researchgate.net/profile/SuvarnsingBhable2>

Google Scholar: <https://scholar.google.co.in/citations?hl=en&user=MIvH-lQAAAAJ>

Github: <https://github.com/suvarnsingbhable>

Declaration

The above information is correct to the best of my knowledge.

Suvarnsing G Bhable.